

RQ1.

The dependent variable was first modelled with a simple model (referred to as the "initial model"). Afterwards, one single predictor (see "added term" column of the model construction table) was added to the baseline model. The model obtained with the added term (i.e. the "compared model") was compared to the same model with the added term (i.e. the "baseline model"). If the compared model led to a significantly better fit with the data (as indicated by the "Pr.Chisq." column), this model was retained and served as the basis for comparison with another augmented model containing an additional added term. On the contrary, if the model with the added term did not significantly improve the data fit, the baseline and more parsimonious model was retained, as indicated in the "comparison output" column. The results of the best-fitting model are reported as the final model.

Initial model: $F0_LOBANOV \sim 0 + (1 | name)[C1 /1/, \text{Hearing typical \& Height Low} = \text{ref. category}]$

Added term	Baseline model	Compared model	npar	AIC	BIC	logLik	deviance	Chisq	Df	Pr.Chisq.	Comparison output	Retained model
Main effect of vowel height	Baseline	M1	4	5679.535	5700.688	-2835.767	5671.535	610.49	2	< 2.2e-16	Better data fit with M1	M1
Random slope of vowel height	M1	M2	6	5683.535	5715.264	-2835.767	5671.535	0	2	1	No better data fit with M2	M1
Main effect of C1	M1	M3	6	5544.633	5576.363	-2766.317	5532.633	138.9	2	< 2.2e-16	Better data fit with M3	M3
Random slope of C1	M3	M4	11	5486.58	5544.751	-2732.29	5464.58	68.053	5	2.60321685026931e-13	Better data fit with M4	M4
Main effect of vowel duration	M4	M5	12	5465.759	5529.218	-2720.88	5441.759	22.821	1	1.77838516939857e-06	Better data fit with M5	M5
Random slope of vowel duration	M5	M6	16	5456.893	5541.505	-2712.447	5424.893	16.866	4	0.00205210661230266	Better data fit with M6	M6
Interaction vowel height - C1	M6	M7	18	5416.0	5511.2	-2690.0	5380.0	44.877	2	1.799e-10	Better data fit with M7	M7
Interaction vowel height - vowel duration	M7	M8	19	5413	5513.4	-2687.5	5375	5.0427	1	0.02473	Better data fit with M8	M8
Interaction vowel duration - C1	M8	M9	21	5416.7	5527.7	-2687.3	5374.7	0.3129	2	0.8552	No better data fit with M9	M8

Final model: $F0_LOBANOV \sim \text{Height} + C1 + \text{vowel duration} + \text{Height:C1} + \text{Height:vowel duration} + (1+C1+\text{vowel duration} | name)[C1 /1/, \text{Hearing typical \& Height Low} = \text{ref. category}]$

	Estimate	Std. error	t-statistic	df	95% CI		p-value
					LL	UL	
(Intercept)	-2.38	0.4	-5.97	557.79	-3.17	-1.6	<0.001
Height High	3.78	0.43	8.86	1101.99	2.94	4.61	<0.001
C1/p/	1	0.21	4.84	151.17	0.6	1.41	<0.001
C1/t/	0.72	0.18	3.92	241.62	0.36	1.08	<0.001
Vowel duration	1.3	1.63	0.79	380.01	-1.91	4.5	0.428
Height High:C1/p/	-0.16	0.2	-0.79	1327.71	-0.55	0.23	0.429
Height High:C1/t/	-1.06	0.24	-4.4	1371.44	-1.53	-0.59	<0.001
Height High:Vowel duration	-4.43	1.95	-2.28	1150.15	-8.25	-0.62	0.023

RQ2.

A. Traditional approach (F0)

Initial model: $F0_LOBANOV \sim 0 + (1 | name)[C1 \ /1/, \text{Hearing typical \& Height Low} = \text{ref. category}]$.

Model construction:

Added term	Baseline model	Compared model	npar	AIC	BIC	logLik	deviance	Chisq	Df	Pr.Chisq.	Comparison output	Retained model
Main effect of vowel height	Baseline	M1	4	5679.535	5700.688	-2835.767	5671.535	610.49	2	< 2.2e-16	Better data fit with M1	M1
Random slope of vowel height	M1	M2	6	5683.535	5715.264	-2835.767	5671.535	0	2	1	No better data fit with M2	M1
Main effect of C1	M1	M3	6	5544.633	5576.363	-2766.317	5532.633	138.9	2	< 2.2e-16	Better data fit with M3	M3
Random slope of C1	M3	M4	11	5486.58	5544.751	-2732.29	5464.58	68.053	5	2.60321685026931e-13	Better data fit with M4	M4
Main effect of vowel duration	M4	M5	12	5465.799	5529.218	-2720.88	5441.799	22.821	1	1.77838516939857e-06	Better data fit with M5	M5
Random slope of vowel duration	M5	M6	16	5456.893	5541.505	-2712.447	5424.893	16.866	4	0.0020321061230266	Better data fit with M6	M6
Main effect of hearing impairment	M6	M7	17	5457.969	5547.869	-2711.985	5423.969	924	1	0.336481328897345	No better data fit with M7	M6
Interaction vowel height - C1	M6	M8	18	5416.016	5511.204	-2690.008	5380.016	43.953	1	3.36289372512627e-11	Better data fit with M8	M8
Interaction vowel height - vowel duration	M8	M9	19	5412.973	5513.45	-2687.487	5374.973	5.043	1	0.024730250593111	Better data fit with M9	M9
Interaction vowel height - hearing impairment	M9	M10	21	5409.556	5520.609	-2683.778	5367.556	7.417	2	0.0245112679943176	Better data fit with M10	M10
Interaction vowel duration - C1	M10	M11	Convergence issues									
Interaction hearing impairment - C1	M10	M12	23	5411.178	5532.808	-2682.589	5365.178	2.3778	2	0.3046	No better data fit with M12	M10
Interaction hearing impairment - vowel duration	M10	M13	22	5411.075	5527.417	-2683.538	5367.075	481	1	0.48816490582095	No better data fit with M13	M10

Final model: $F0_LOBANOV \sim \text{Height} + C1 + \text{vowel duration} + \text{Height:C1} + \text{Height:vowel duration} + \text{Height:Hearing impairment} + (1 + C1 + \text{vowel duration} | name); C1 \ /1/, \text{Hearing typical \& Height Low} = \text{ref. category}.$

	Estimate	Std. error	df	t-statistic	Bootstrapped 95% CI		p-value
					LL	UL	
Intercept	-2.46	0.40	573.46	-6.17	-3.55	-0.12	<0.001
Height High	3.86	0.43	1112.48	9.06	1.30	4.85	<0.001
C1/p/	1.00	0.21	151.85	4.85	0.42	2.03	<0.001
C1/t/	0.71	0.18	246.91	3.89	0.06	1.35	<0.001
Vowel duration	1.30	1.63	408.29	0.80	-9.03	5.70	0.425
Height High:C1/p/	-0.15	0.20	1327.43	-0.77	-0.91	0.51	0.439
Height High:C1/t/	-1.06	0.24	1370.89	-4.42	-2.19	-0.25	<0.001
Height High:vowel duration	-4.39	1.94	1180.31	-2.26	-8.77	6.97	0.024
Height Low:Hearing impaired	0.48	0.18	705.08	2.65	-0.07	0.98	0.008
Height High: Hearing impaired	-0.10	0.14	156.19	-0.67	-0.52	0.35	0.505

B. Control for differences in degree of opening (F1-F0)

Initial model: $Distance_F1_LOBANOV_F0_LOBANOV \sim 0 + (1 | name)[C1 \ /1/, \text{Hearing typical \& Height Low} = \text{ref. category}]$.

Model construction:

Added term	Baseline model	Compared model	npar	AIC	BIC	logLik	deviance	Chisq	Df	Pr.Chisq.	Comparison output	Retained model
Main effect of vowel height	Baseline	M1	4	5885.715	5906.868	-2938.857	5877.715	1657.945	2	< 2.22e-16	Better data fit with M1	M1
Random slope of vowel height	M1	M2	6	5885.921	5917.65	-2936.96	5873.921	3.794	2	0.149986759419635	No better data fit with M2	M1
Main effect of C1	M1	M3	6	5778.621	5810.351	-2883.311	5766.621	111.09	2	< 2.22e-16	Better data fit with M3	M3
Random slope of C1	M3	M4	11	5726.6	5794.77	-2852.3	5704.6	62.022	5	4.64117456017871e-12	Better data fit with M4	M4
Main effect of vowel duration	M4	M5	12	5680.01	5743.468	-2828.005	5656.01	48.59	1	3.15449865822173e-12	Better data fit with M5	M5
Random slope of vowel duration	M5	M6	16	5678.313	5762.924	-2823.156	5646.313	9.697	4	0.0458528719146359	Better data fit with M6	M6
Main effect of hearing impairment	M6	M7	17	5681.485	5771.386	-2823.743	5647.485	0	1	1	No better data fit with M7	M6
Interaction vowel height - C1	M6	M8	18	5627.992	5723.181	-2795.996	5591.993	55.493	1	9.28013311809079e-14	Better data fit with M8	M8
Interaction vowel height - vowel duration	M8	M9	19	5629.014	5729.491	-2795.507	5591.014	978	1	0.322605134573184	No better data fit with M9	M8
Interaction vowel height - hearing impairment	M8	M10	20	5624.491	5730.256	-2792.246	5584.491	6.523	1	0.010649367996953	Better data fit with M10	M10
Interaction hearing impairment - C1	M10	M11	22	5627.772	5744.114	-2791.886	5583.772	719	2	0.697957310403714	No better data fit with M11	M10
Interaction hearing impairment - vowel duration	M10	M12	22	5632.775	5749.116	-2794.387	5588.775	0	2	1	No better data fit with M12	M10
Interaction hearing impairment - vowel duration	M10	M13	21	5626.6	5737.7	-2792.6	5584.6	0	1	1	No better data fit with M13	M10

Final model: $Distance_F1_LOBANOV_F0_LOBANOV \sim Height + C1 + vowel\ duration + Height:C1 + Height: Hearing\ impairment + (1+C1+vowel\ duration | name)[C1\ /1/, Hearing\ typical \& Height\ Low = ref.\ category]$.

	Estimate	Std. error	df	t-statistic	Bootstrapped 95% CI		p-value
					LL	UL	
Intercept	3.76	0.30	328.98	12.34	2.04	4.97	<0.001
Height high	-6.08	0.18	1334.72	-33.09	-6.79	-5.02	<0.001
C1/p/	-1.02	0.23	140.05	-4.51	-2.02	0.00	<0.001
C1/t/	-1.07	0.19	196.50	-5.68	-1.84	-0.08	<0.001
Vowel duration	2.50	1.21	169.14	2.07	-2.49	8.84	0.040
Height high: C1/p/	0.29	0.21	1310.05	1.37	-0.61	0.94	0.171
Height high: C1/t/	1.66	0.23	1370.37	7.34	0.38	2.73	<0.001
Height low: HI	-0.55	0.23	157.02	-2.38	-1.40	0.34	0.018
Height high: HI	0.13	0.16	94.85	0.82	-0.35	0.65	0.413

RQ3.

A. Traditional approach (F0)

Initial model: $F0_LOBANOV \sim 0 + (1 | name)[C1\ /1/, Hearing\ typical \& Height\ Low = ref.\ category]$.

Model construction:

Added term	Baseline model	Compared model	npar	AIC	BIC	logLik	deviance	Chisq	Df	Pr. Chisq.	Comparison output	Retained model
Main effect of vowel height	Baseline	M1	4	5679.535	5700.688	-2835.767	5671.535	598.72599	1	< 2.22e-16	Better data fit with M1	M1
Random slope of vowel height	M1	M2	6	5683.535	5715.264	-2835.767	5671.535	0	2	1	No better data fit with M2	M1
Main effect of C1	M1	M3	6	5544.633	5576.363	-2766.317	5532.633	138.901	2	< 2.22e-16	Better data fit with M3	M3
Random slope of C1	M3	M4	11	5486.5800	5544.7507	-2732.2900	5464.5800	68.05328	5	2.4032e-13	Better data fit with M4	M4
Main effect of vowel duration	M4	M5	7	5526.606	5563.623	-2756.303	5512.606	20.028	1	7.63305731339011e-06	Better data fit with M5	M5
Random slope of vowel duration	M5	M6	9	5510.34	5557.934	-2746.17	5492.34	20.265	2	3.9759160035469e-05	Better data fit with M6	M6
Main effect of hearing impairment	M6	M7	Convergence issues									
Interaction vowel height - C1	M6	M8	18	5416.016	5511.204	-2690.008	5380.016	44.87718	2	1.7991e-10	No better data fit with M7	M6
Interaction vowel height - vowel duration	M8	M9	19	5412.973	5513.45	-2687.487	5374.973	5.043	1	0.0247302505935111	Better data fit with M8	M8
Interaction vowel height - hearing impairment	M9	M10	23	5412.716	5534.345	-2683.358	5366.716	8.2574	4	0.082592	No better data fit with M10	M9
Interaction vowel duration - C1	M9	M11	21	5416.66	5527.713	-2687.33	5374.66	0.31293	2	0.85516	No better data fit with M11	M9
Interaction hearing impairment - C1	M9	M12	25	5417.476	5549.682	-2683.378	5367.476	7.49703	6	0.27731	No better data fit with M12	M9
Interaction hearing impairment - vowel duration	M9	M13	21	5418.083	5529.137	-2688.042	5376.083	0	2	1	No better data fit with M13	M9

Final model: $F0_LOBANOV \sim Height + C1 + vowel\ duration + Height:C1 + Height:vowel\ duration + (1+C1+vowel\ duration | name)[C1\ /1/, Hearing\ typical \& Height\ Low = ref.\ category]$.

	Estimate	Std. error	df	t-statistic	95% CI		p-value
					LL	UL	
(Intercept)	-2.38	0.4	557.79	-5.97	-3.17	-1.6	<0.001
Height High	3.78	0.43	1101.99	8.86	2.94	4.61	<0.001
C1/p/	1	0.21	151.17	4.84	0.6	1.41	<0.001
C1/t/	0.72	0.18	241.62	3.92	0.36	1.08	<0.001
Vowel duration	1.3	1.63	380.01	0.79	-1.91	4.5	0.428
Height High:C1/p/	-0.16	0.2	1327.71	-0.79	-0.55	0.23	0.429
Height High:C1/t/	-1.06	0.24	1371.44	-4.4	-1.53	-0.59	<0.001
Height High:Vowel duration	-4.43	1.95	1150.15	-2.28	-8.25	-0.62	0.023

B. Control for differences in degree of opening (F1-F0)

Initial model: $Distance_F1_LOBANOV_F0_LOBANOV \sim 0 + (1 | name)[C1 /1/, Hearing\ typical \& Height\ Low = ref.\ category]$.

Model construction:

Added term	Baseline model	Compared model	npar	AIC	BIC	logLik	deviance	Chisq	Df	Pr.>Chisq.	Comparison output	Retained model
Main effect of vowel height	Baseline	M1	4	5885.715	5906.868	-2938.857	5877.715	1657.945	2	< 2.22e-16	Better data fit with M1	M1
Random slope of vowel height	M1	M2	6	5885.921	5917.65	-2936.96	5873.921	3.794	2	0.149986759419635	No better data fit with M2	M1
Main effect of C1	M1	M3	6	5778.621	5810.351	-2883.311	5766.621	111.09	2	< 2.22e-16	Better data fit with M3	M3
Random slope of C1	M3	M4	11	5726.6	5784.77	-2852.3	5704.6	62.022	5	4.64117456017871e-12	Better data fit with M4	M4
Main effect of vowel duration	M4	M5	12	5680.01	5743.468	-2828.005	5656.01	48.59	1	3.15449865822173e-12	Better data fit with M5	M5
Random slope of vowel duration	M5	M6	16	5678.313	5762.924	-2823.156	5646.313	9.697	4	0.0458528719146359	Better data fit with M6	M6
Main effect of hearing impairment	M6	M7	18	5685.269	5780.458	-2824.635	5649.269	0	2	1	No better data fit with M7	M6
Interaction vowel height - C1	M6	M8	18	5627.993	5723.181	-2795.996	5591.993	54.32	2	1.602e-12	Better data fit with M8	M8
Interaction vowel height - vowel duration	M8	M9	19	5629.014	5729.491	-2795.507	5591.014	0.9784	1	0.3226	No better data fit with M9	M8
Interaction vowel height - hearing impairment	M8	M10	22	5627.852	5744.193	-2791.926	5583.852	8.1409	4	0.8655	No better data fit with M10	M8
Interaction hearing impairment - C1	M8	M11	24	5638.296	5765.214	-2795.148	5590.296	1.6967	6	0.9454	No better data fit with M11	M8
Interaction C1 - vowel duration	M8	M12	20	5631.462	5737.227	-2795.731	5591.462	0.5308	2	0.7669	No better data fit with M12	M8
Interaction hearing impairment - vowel duration	M8	M13	24	5629	5735.9	-2790.5	5581	10.986	6	0.08883	No better data fit with M13	M8

Final model: $Height + C1 + vowel\ duration + Height:C1 + (1+C1+vowel\ duration | name)[C1 /1/, Hearing\ typical \& Height\ Low = ref.\ category]$.

	Estimate	Std. error	df	t-statistic	95% CI		p-value
					LL	UL	
(Intercept)	3.65	0.3	321.78	12.07	3.05	4.24	<0.001
Height High	-5.96	0.18	1375.35	-33.26	-6.31	-5.61	<0.001
C1/p/	-1.03	0.23	140.89	-4.51	-1.47	-0.58	<0.001
C1/t/	-1.08	0.19	190.64	-5.68	-1.45	-0.71	<0.001
total_duration	2.61	1.21	158.84	2.15	0.23	4.98	0.033
Height High:C1/p/	0.3	0.21	1309.03	1.39	-0.12	0.72	0.165
Height High:C1/t/	1.65	0.23	1369.12	7.28	1.21	2.1	<0.001

Bibliography